Plumbing and wet-room study · round 1 · inventory, overlay, constraints, scores, alternatives, three scenarios, the utility room · Taif / Palm House V8.1 · STUDY ONLY

Design review E-03, G-07 · open item 10 · 2 Nov 2026 · the latest geometry (the flipped stair, the family bath F-b, the south wing v2, the maid room and kitchen changes, the site plan's manholes and septic route) · **every size is assumed, to be designed by the plumbing engineer** · nothing in the design files changes until the owner decides

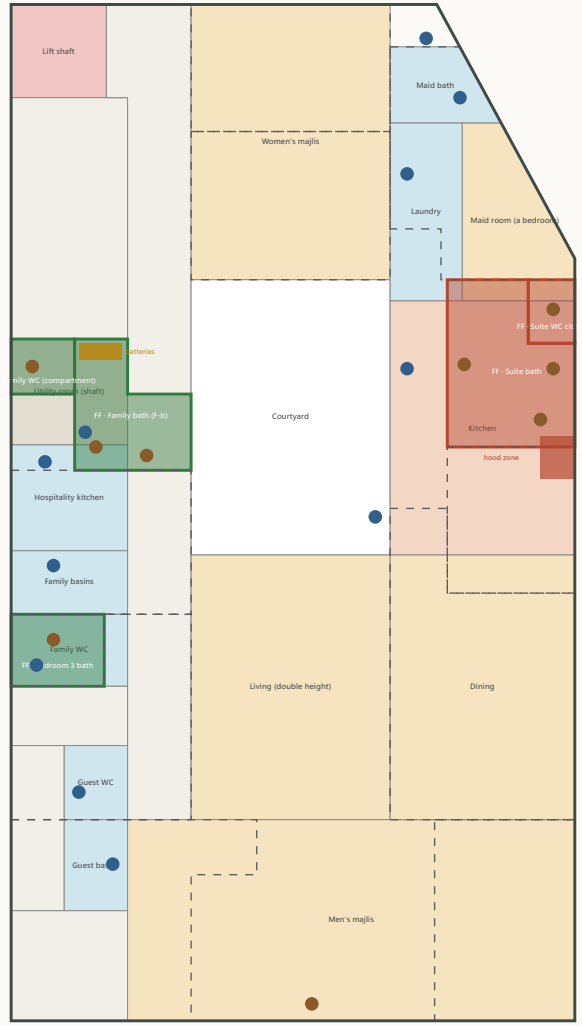
Principles applied in every scenario (owner's goals: maintenance without breaking floors or walls, no smells, low leak risk, quality, controlled cost)

1. **Two clusters only:** west round the shaft (the utility room, the hospitality kitchen, the family basins and WC, the guest WC and basins; above: the family bath F-b and its WC compartment, bedroom 3's bath) and east along the east wall (the kitchen, the laundry, the maid bath; above: the suite bath and its WC closet). **No third cluster** (bedroom 1 stays without a bath: decided).
2. **Ground floor:** the drains buried under the ground slab, a **cleanout at every change of direction**, an **inspection chamber outside each cluster** (the west manhole on the driveway, the east manhole in the yard), reachable from above; then the collector manhole and the existing septic (site plan).
3. **Stacks:** vertical stacks inside the **west shaft** (doors on each floor) or an **accessible duct** (scenario C: an internal east duct); acoustic (triple-layer) pipes on every stack; the east stack today runs outside the east wall behind a removable cladding.
4. **First-floor bathrooms:** a **sunken waterproofed slab** (0.10–0.15 m, the leak kept inside the bathroom) wherever the bay allows it (the ribbed bays; the suite bath has it, X10), OR suspended drains only above **service rooms and corridors** behind access panels with acoustic lagging; **never** above bedrooms, the majlis, the living or the kitchen's hood zone.
5. **No smell:** real **vent stacks to the roof** (the west shaft and the east stack; ≥ 3 m from the HRV intake and the malqaf), **deep-seal traps** (≥ 75 mm) on every fixture, **anti-odour floor drains** (membrane or ball type, with a primer) on the rarely used drains: the courtyard, the suite court, the laundry and the utility room, the breeze channel's sump.
6. **Water:** a **manifold system** (one continuous PEX pipe from the manifold to each fixture, no buried joints), a **manifold cabinet per floor near the shaft** with a valve per fixture (west: the utility room GF, the family bath's wall FF; east: the laundry GF, the suite WC closet's wall FF); **one water heater per cluster** close to its bathrooms (west: the utility room; east: the laundry, serving the kitchen, the maid bath and the suite bath above); a **booster pump** from the tank (site plan) with a main sediment filter in the utility room; an **RO unit** at the kitchen sink.
7. **Leak protection:** leak sensors under the heaters, the washer, the dishwasher and every basin cabinet, wired to an **automatic main shut-off valve** after the filter; drip trays under the heaters and the washer, drained to a floor drain; the first-floor bathrooms' sunken slabs are the second line.

1 · Inventory: every wet point (both floors and the roof)

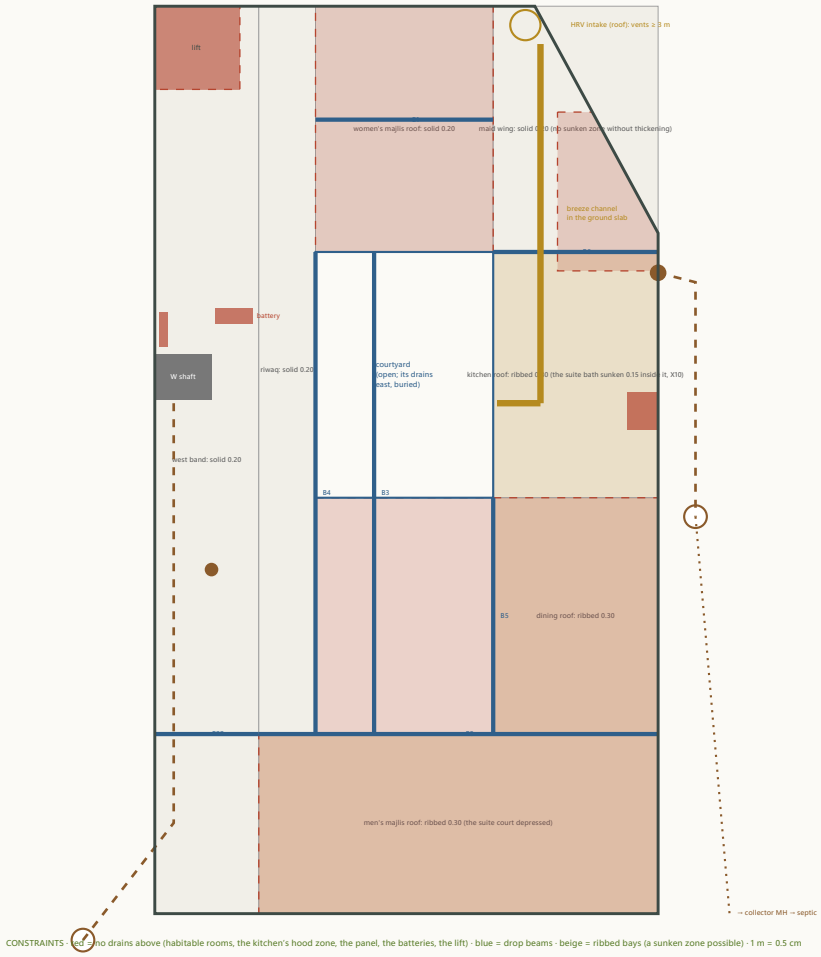
Fl.	Room	Fixtures	Drains to	Branch m	Use
GF	Hospitality kitchen	sink	W-RUN	0.3	daily
GF	Family basins	2 basins (wudu)	W-RUN	0.5	daily
GF	Family WC	WC + basin	W-RUN	0.1	daily
GF	Guest WC	WC + basin	W-RUN	1.1	guests
GF	Guest basins	trough, 3 taps	W-RUN	3.2	guests
GF	Utility room	heater drip tray · condensate	W-SHAFT	0.3	rare
GF	Kitchen	sink + dishwasher	E-MH	9	daily
GF	Laundry	washer · dryer · utility sink · floor drain	E-MH	13.5	daily / rare (floor drain)
GF	Maid bath	WC · basin · shower	E-MH	13.2	daily
GF	Courtyard	step drain · overflow · basin + planter	RAIN	6.4	rain / washing
GF	Air tower	breeze channel sump · HRV condensate	E-MH	14.5	rare
FF	Family bath (F-b)	shower	W-SHAFT	0.6	daily
FF	Family bath (F-b)	double basin	W-SHAFT	1.7	daily
FF	Family WC (compartment)	WC	W-SHAFT	1.2	daily
FF	Bedroom 3 bath	shower · WC · basin	FWC	0.5	daily
FF	Suite WC closet	WC	E-STACK	0.5	daily
FF	Suite bath	shower (east wall)	E-STACK	1.6	daily
FF	Suite bath	tub (by the east window)	E-STACK	3.1	weekly
FF	Suite bath	double vanity (west wall)	E-STACK	3	daily
FF	Suite court	3 linear drains · box gutter · overflow	RAIN	3.2	rain
roof	Roof	11 outlets → internal downpipes	RAIN	—	rain
roof	Roof (plant)	AC condensate (Z2b) · HRV condensate (base)	E-STACK	—	daily (summer)

Branch = the horizontal drain from the fixture to its collector (the buried run, a stack or the manhole), measured on the routes drawn; the roof outlets (11) and the AC / HRV condensate are listed, not scored. Total of the scored branches: 77.5 m.



OVERLAY · the first-floor wet rooms (green = over a wet / service room, red = over the kitchen) on the ground floor · dots: wet points (blue GF, brown FF) · 1 m = 0.56 cm

2 · Constraints map · what lies under each first-floor wet room · the stacks and the routes to the septic



CONSTRAINTS · red = no drains above (habitable rooms, the kitchen's hood zone, the panel, the batteries, the lift) · blue = drop beams · beige = ribbed bays (a sunken zone possible) · 1 m = 0.5 cm

No-drain zones above

- **Habitable rooms (no drains in their ceilings):** the women's majlis, the men's majlis, the living and the dining, the maid room (a bedroom); on the first floor every bedroom, the kids' room, the family living, the office and the gym carry only the roof (no first-floor fixtures above them). **The kitchen:** its **hood zone** on the east wall (y 10.2–11.2) takes no drain above it; the rest of the kitchen's ceiling carries the suite bath only inside a sunken tanked slab (no suspended pipes in the kitchen's ceiling). **The electrical panel and the battery space** (the utility room's north part) and the **lift shaft**: nothing above or beside.

Structure

- **Drop beams** (blue): B1 (y 3.0), B4 (x 4.25), B5 (x 8.95 south of y 13), B6 (y 6.5), B8 / B22 (y 19.25), the edge beam B3: a drain never crosses a drop beam; branches run parallel to them or through sleeves cast in the hidden beams (the engineer).
- **Slabs:** the west band, the riwaq, the maid wing and the women's majlis roof are **solid 0.20**: a sunken bathroom zone there means a thicker bay (0.35 with the top depressed 0.15) or a raised floor; the kitchen, dining and men's majlis roofs are **ribbed 0.30**: a 0.10–0.15 depressed zone is possible (the suite bath has one). The family bath F-b over the utility room / hospitality kitchen is on the solid west band: its sunken zone needs the thicker bay (X10-type detail) or its drains stay suspended above the two **service** rooms behind panels (allowed by principle 4).
- **The malqaf breeze channel** (amber) in the ground slab from the tower south on x 10.2 to y 10.5 then west under the kitchen's sink counter: the kitchen's drain crosses it (the sink's branch passes **under** the channel's base slab, or the channel bridges it: the engineers); the **foundations** (pads ≈ 1.5 × 1.5 under the columns, strip footings under the RC walls): the buried drains pass between them; the **courtyard**: its drains run east under the kitchen floor to the east manhole's rain branch.

The two stacks and the routes to the septic

- **West:** the shaft (0–1.5 × 9.2–10.4) carries the west stack and vent from the family bath / WC compartment down to the buried west run, which collects the hospitality kitchen, the family basins, the family WC (its own small stack takes bedroom 3's bath from above) and the guest WC and basins, and leaves under the west wall to the **west manhole** on the driveway. **East:** the stack outside the east wall beside the suite WC closet takes the suite bath, drops to the yard and runs south to the **east manhole**, which also takes the kitchen, the laundry, the maid bath (buried, east) and the courtyard's rain branch. Both manholes run to the **collector manhole** at the garden's south-east and the **existing septic** (site plan; the capacity, the condition and the well distance flagged).

Overlay verdicts

First-floor wet room	Below it	Verdict
Family bath (F-b)	Utility room (shaft), Hospitality kitchen, Lobby + riwaq	good: over wet / service
Family WC (compartment)	Utility room (shaft)	good: over wet / service
Bedroom 3 bath	Family WC	good: over wet / service
Suite bath	Maid room (a bedroom), Laundry, Kitchen	problem: over the kitchen
Suite WC closet	Maid room (a bedroom), Kitchen	problem: over the kitchen

3 · Score of every wet room · 4 · Alternatives for the rooms that score poorly

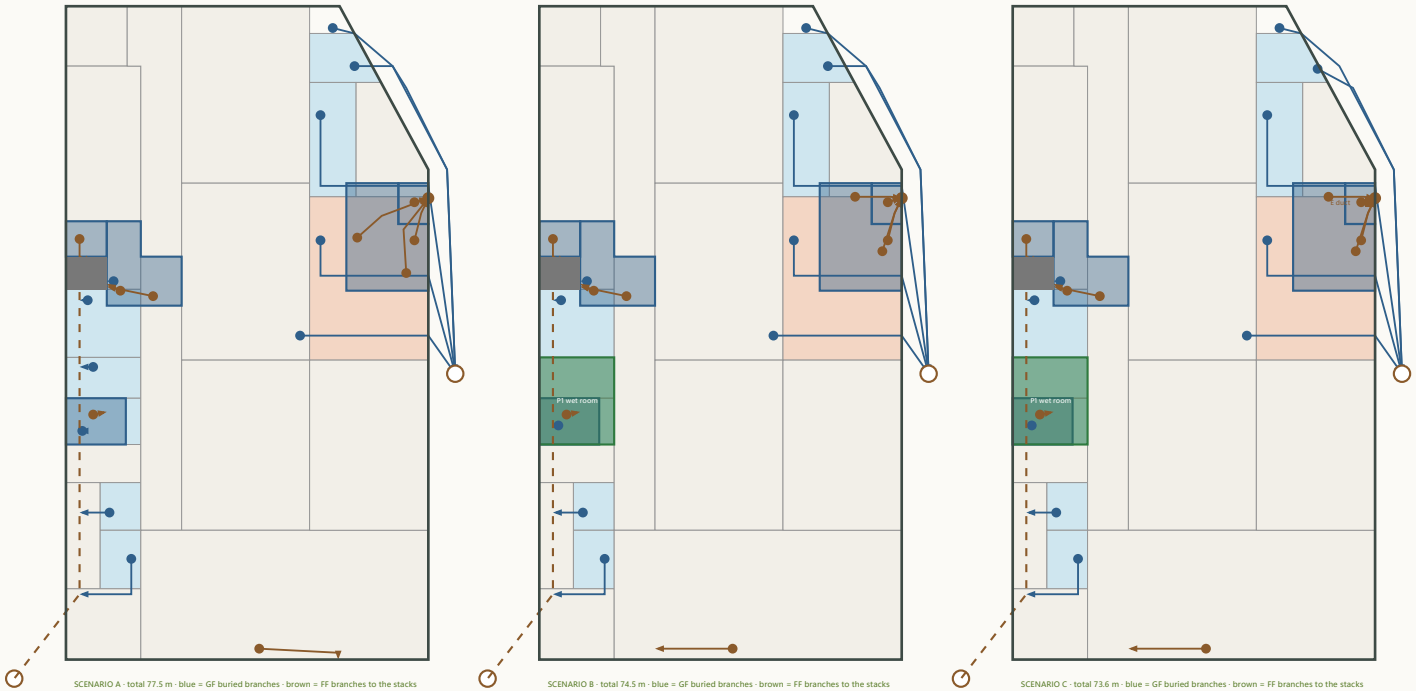
Fl.	Room	Fixture	m	bends	over	access	smell	score / 10
GF	Hospitality kitchen	sink	0.3	0	buried	cleanout	trap + vent	9.9
GF	Family basins	2 basins (wudu)	0.5	0	buried	cleanout	trap + vent	9.8
GF	Family WC	WC + basin	0.1	0	buried	cleanout	trap + vent	10
GF	Guest WC	WC + basin	1.1	0	buried	cleanout	trap + vent	9.5
GF	Guest basins	trough, 3 taps	3.2	1	buried	cleanout	trap + vent	7.7
GF	Utility room	heater drip tray · condensate	0.3	0	buried	shaft door	anti-odour drain	8.9
GF	Kitchen	sink + dishwasher	9	2	buried	cleanout	trap + vent	5.6
GF	Laundry	washer · dryer · utility sink · floor drain	13.5	2	buried	cleanout	anti-odour drain	4.6
GF	Maid bath	WC · basin · shower	13.2	3	buried	cleanout	trap + vent	5
GF	Courtyard	step drain · overflow · basin + planter	6.4	1	buried	cleanout	anti-odour drain	5.3
GF	Air tower	breeze channel sump · HRV condensate	14.5	3	buried	cleanout	anti-odour drain	4
FF	Family bath (F-b)	shower	0.6	0	wet / service	shaft door	trap + vent	9.7
FF	Family bath (F-b)	double basin	1.7	0	wet / service	shaft door	trap + vent	9.1
FF	Family WC (compartment)	WC	1.2	1	wet / service	shaft door	trap + vent	8.7
FF	Bedroom 3 bath	shower · WC · basin	0.5	0	wet / service	stack panel	trap + vent	9.7
FF	Suite WC closet	WC	0.5	0	kitchen	outside stack	trap + vent	6.7
FF	Suite bath	shower (east wall)	1.6	1	kitchen	outside stack	trap + vent	5.5
FF	Suite bath	tub (by the east window)	3.1	1	kitchen	outside stack	trap + vent	4.8
FF	Suite bath	double vanity (west wall)	3	2	kitchen	outside stack	trap + vent	4.1
FF	Suite court	3 linear drains · box gutter · overflow	3.2	1	majlis roof (rain)	outside stack	anti-odour drain	4.7

Score = branch length (3) + bends (2) + stacked over a wet / service room (2) + maintenance access (1.5) + smell risk (1.5); structural impact and cost are in the text; the weights are this study's, not a code. Mean today: 7.2.

The rooms that score poorly, and their alternatives

- **The suite bath and its WC closet (over the kitchen):** the only first-floor wet room over a problem room, and the house's longest first-floor branch (the vanity on the west wall, 3 m). **Alternatives:** (a) **keep it** with its sunken tanked slab 0.15 (exists, X10), every branch inside the sunken zone, the vanity branch along the north wall; nothing in the kitchen's ceiling; (b) **move the vanity to the east side** beside the stack (2.1 m) and the tub to the north wall (1.7 m): the branches shorten, the room keeps its function (a double vanity still faces the east light from the window); (c) **swap the bath and the dressing** (the bath over the kitchen's south part / the dining): still over the kitchen and farther from the stack: worse; (d) **move the bath over the laundry / maid wing** ($y \le 7.0$): the suite lobby and bedroom 1 are there: not possible without re-planning the suite. **Chosen for B and C: (b).**
- **The suite court's gutter outlets** (rain, in the men's majlis's south wall): (a) keep, with acoustic lagging; (b) run the two outlets to the court's ends and down inside the **gym's and the office's** walls to the garden: no pipe in a majlis wall. **Chosen for B: (b).**
- **The courtyard's drain, overflow and basin** (6.4 m east under the kitchen floor): buried, fine; anti-odour drain and the basin's own trap; a cleanout at the kitchen's west wall. Keep.
- **The maid bath** (13.2 m round the cut corner): (a) keep; (b) **align its fixtures on the cut wall** with one short branch out through the cut wall to the strip, and the breeze channel's sump and the HRV condensate into the same branch. **Chosen for C.**
- **The ground-floor shower** (G-07, deferred to this study): **P1** (the family WC + basins as one wet room on the west run, 0.5–1.5 m branches, a level floor drain in the ground slab: easy) scores as well as any fixture here; **P2** (the hospitality kitchen, 1 m from the shaft) is the plumbing optimum but re-plans three rooms; **P1-lite** (the family WC room only) is P1 without the basins. **Chosen for B and C: P1**, for its function as much as its plumbing.
- **The laundry:** fine (buried east); C adds the east heater and manifold cabinet beside the washer and moves the utility sink next to it (one trap group).

4 · Three whole-house scenarios · A minimum · B moderate · C ideal · the comparison and the recommendation



Criterion	A · minimum	B · moderate	C · ideal
What changes	nothing moves; cleanouts, two vent stacks, anti-odour drains, acoustic stacks, the manifold system, two heaters, leak protection	A + the GF shower P1 (the family WC + basins → one wet room), the suite vanity beside the stack and the tub on the north wall, the court's outlets down the gym / office walls	B + an internal accessible east duct (0.35 × 0.35, the kitchen's NE corner under the suite WC closet) with doors on both floors; the maid bath on the cut wall; the east heater + manifold in the laundry
Total horizontal drain (scored branches)	77.5 m	74.5 m	73.6 m
Longest first-floor foul branch	3.1 m (the suite vanity)	2.1 m (the suite vanity)	2.1 m
Wet rooms over a problem room	3 (the suite bath over the kitchen: sunken tanked slab)	3 (the same)	3 (the same; its branches now reach an internal duct)
Maintenance access	the west shaft (doors), the east stack outside behind cladding, cleanouts	as A; the GF wet room's floor drain from above	every stack behind a door on each floor; no outside pipe
Smell risk	low: vents, deep traps, anti-odour drains	low	lowest: the rare drains grouped on the east duct with a primer
Structural changes	none (the suite bath's sunken zone exists)	B14 under the removed wall of the family WC (a drop beam, the shower study)	B + a 0.35 × 0.35 opening through the kitchen roof slab beside C33 and the east wall (the engineer)
Rough relative cost	1.0 (the base: pipes, stacks, manifolds, 2 heaters, sensors)	≈ 1.15 (one wet room re-fitted, a beam, a window)	≈ 1.3 (the duct, the maid bath re-fit, the laundry re-fit)
Function impact	none	the wudu basins behind a sliding door; the suite vanity on the east side (lit by the window)	B + the maid bath re-planned (same size), the laundry loses 0.6 m of counter
Fixtures scoring below 5 / mean	5 of 20 / 7.2	2 of 19 / 7.2	2 of 19 / 7.2

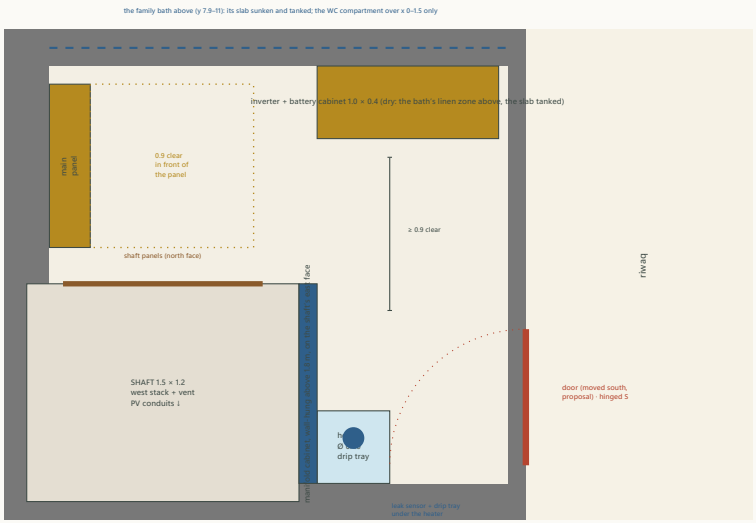
Recommendation: B, plus C's internal east duct if the engineer allows the slab opening

- The house is already well clustered: every first-floor wet room but one stands over a wet or service room, and the one exception (the suite bath over the kitchen) is mitigated by its sunken tanked slab and cannot be moved without re-planning the suite. So the gains lie in the **routes, the access and the venting** (A) and in a few short moves (B): the GF shower as P1, the suite vanity beside the stack, the court's outlets out of the majlis wall. B costs little and removes the longest branch.
- C's one real gain is the **internal east duct**: it brings the east stack indoors behind a door on each floor (maintenance without breaking walls, no pipe on the facade) at the price of a slab opening in the kitchen's corner; the maid bath and laundry re-fits are small. **Take the duct if the engineer accepts the opening; leave the rest of C.**
- **In every scenario:** the sunken slabs are the second line against leaks; the manifold system and the sensors with the automatic valve are the first; the two vent stacks run to the roof ≥ 3 m from the HRV intake (12.5 / 7.4 m).

Decisions needed from the owner

1. **Scenario:** A, B (recommended) or C; and, with B, **the internal east duct** from C (yes / no, subject to the engineer).
2. **The GF shower:** P1 confirmed as the wet room of the west cluster (or P1-lite / P2)?
3. **The suite bath:** the vanity moved to the east side beside the stack (B) — accepted?
4. **The suite court's outlets** down the gym / office walls instead of the majlis wall (B)?
5. **Heaters:** two electric cylinders (west ≈ 100–120 L in the utility room, east ≈ 150 L in the laundry), or a central heat-pump unit on the roof with a ring main (less space inside, more pipe)? (to be sized by the engineer)
6. **The utility room's door** moved south (y 9.45–10.2, 0.75 wide) to free the north wall for the panel and the batteries (page 5)?

The utility room (west cluster): the shaft, the panel, the battery space, the heater, the manifold, the access clearances · the water kept away from the batteries



UTILITY ROOM · 2.525 × 2.3 clear · 1 m = 2.4 cm · north up

Layout (proposal, every size assumed)

- **The room:** x 0–2.75, y 7.9–10.4 on axes, **2.525 × 2.3 m clear**; the **shaft** 1.5 × 1.2 in its south-west corner (the west stack and vent, the PV conduits from the roof, the cold and hot risers to the first floor), its **access panels on the north face** (removable, 1.1 m wide) so the shaft is serviced from the room's north part; the **door** moved from y 8.2–9.0 to **y 9.45–10.2** (0.75 wide, hinged on the south jamb, opening west: its swing stays south of the battery wall), a proposal for `design_gf.typ`.
- **Dry side (north and west):** the **main electrical panel** on the west wall (x 0.125–0.35, y 8.1–9.0) with 0.9 m clear in front; the **inverter and the battery cabinet** 1.0 × 0.4 on the **north wall's east part** (x 1.6–2.6): nothing wet above or beside it — above it lies the family bath's **dry linen zone** (x 1.5–2.75, y 7.9–9.2), the WC compartment sits over x 0–1.5 only, and the bath's slab is sunken and tanked; the heater is 1.5 m away, the manifold 0.8 m, with a drip tray under each. The alarm panel and the recorder (site plan) share the dry side.
- **Wet side (south):** the **water heater** (an electric cylinder Ø 0.40, ≈ 100–120 L, floor-standing on a drip tray with a leak sensor) at x 1.6–2.0 beside the shaft's east face; the **manifold cabinet** wall-hung above 1.8 m on the shaft's east face (x 1.5–1.6, y 9.2–10.3, above the heater's top) with the valves, the main sediment filter, the booster pump's connection and the automatic shut-off valve; the floor drain by the heater into the west stack.
- **Clearances:** 0.9 m in front of the panel **OK**, the battery's front free to the door swing **OK**, the heater reachable from the room's middle, the shaft panels reachable from the north part; the free floor ≈ 4.3 m².

Flags

- **The room is small:** with the shaft it leaves 4.3 m² of free floor; a 200 L cylinder or a heat-pump unit does not fit (a roof heat pump would); the batteries and the water share one room by the owner's decision, with the layout above as the condition. **The electrical engineer confirms** the battery cabinet's space, its ventilation (a lithium cabinet, ventilated to the riwaq or the shaft), its fire safety (a steel cabinet, ≥ 0.5 m from combustibles, the smoke alarm by the stair, ideally a 60-minute separation to the stair store) and the panel's position.
- **If the engineer wants more:** the store under the stair's half landing (north of this wall, 2.525 × 1.25, 1.65 m high, the stair above it, dry) can take the battery cabinet instead, with a fire-rated door; the owner's decision today is the utility room.

Study checks passed (9)

Checks: every first-floor wet room except the suite bath stands over a wet or service room (the overlay) · the suite bath and its WC closet stand over the kitchen (reported as the one unavoidable problem, mitigated by the sunken tanked slab) · no first-floor drain crosses the kitchen's hood zone (the tub's branch kept north of y 10.2 at the east wall) · the two vent stacks ≥ 3 m from the HRV intake (west 12.5 m, east 7.4 m) · every first-floor foul branch ≤ 4 m horizontal (max 3.1 m, the suite vanity today) · scenario B shortens the longest first-floor foul branch to 2.1 m and the total to 74.5 m (A: 77.5), and leaves fewer fixtures below 5 (2 vs 5) · utility room: the battery cabinet not under the WC compartment (x ≥ 1.5), ≥ 1 m from the heater (1.5) and ≥ 0.75 m from the manifold (0.8) · utility room: the door's swing clear of the heater, the battery and the manifold; 0.9 m in front of the panel; the shaft panels on its north face reachable · utility room: free floor ≈ 4.3 m² (small: reported). Every size is assumed, to be designed by the plumbing engineer; nothing here changes the room CSVs or the design files.